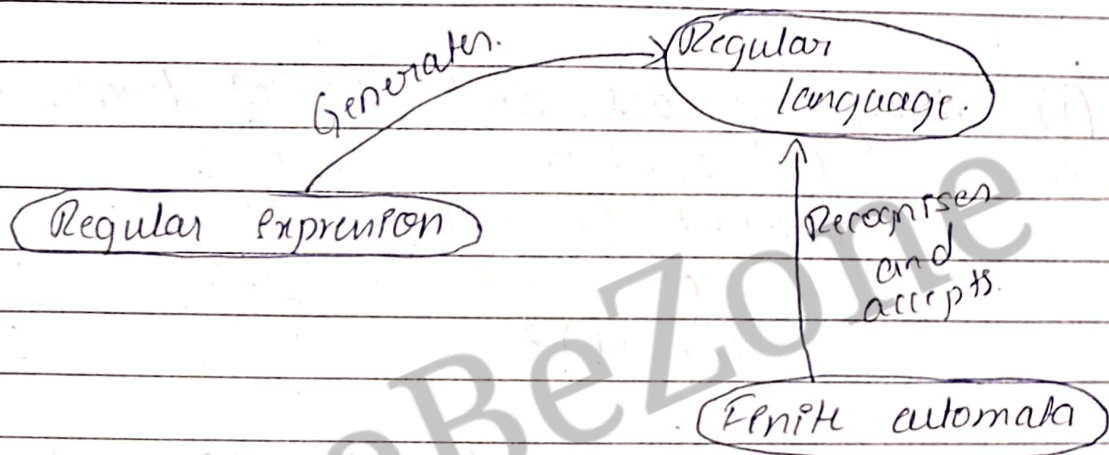


Module - 02.REGULAR EXPRESSION

→ Relationship b/w Regular expression, Regular languages and finite automata.



* The operators of Regular expression.

→ The operations on languages are:-

- (1) Union ($\cup$ or $+$)
- (2) Concatenation ($\cdot$)
- (3) Closure (closure $*$)
- (4) Closure plus ($+$)

→ Formal definition of Regular expression.

→ The basis consists of three parts.

- (1) The constants ϵ and ϕ are regular expressions, denoting the languages $\{\epsilon\}$ and ϕ respectively. That is $L(\epsilon) = L\{\epsilon\}$, and $L(\phi) = \phi$
- (2) If a is any symbol then a is a regular expression. This expression denotes.

the language $\{a^n \mid n \text{ is even}\} = \{aa\}^*$

(3) A variable capitalization and order such as a variable representing any language.

* Write the regular expression for the following

(1) Strings of a's and b's of length ≤ 2 .

$$\{ \epsilon, a, b, aa, ab, ba, bb \}$$

$$\text{OR } (\epsilon + a + b)^2$$

(2) Strings of a's and b's of length ≤ 3

$$\{ \epsilon + a + b \}^3$$

(3) Strings of a's and b's having even length.

$$\{ \epsilon, aa, ab, ba, bb, \}$$

$$(a+b)(a+b)^*$$

OR

$$(aa+ab+ba+bb)^*$$

(4) Strings of a's and b's having odd length

$$(a+b)((a+b)(a+b))^*$$

OR

$$((a+b)(a+b))^* (a+b)$$

(5) $\{0,1\}^*$ with atleast 3 consecutive zeros

$$(0+1)^* 000 (0+1)^*$$

(6) Starting with a's and ending with b

$$a(a+b)^* b$$

(7) Strings of a's and b's whose second symbol from right end is a

$$(a+b)^* a(a+b)$$

(8) Third symbol from the right hand is a

$$(a+b)^* a(a+b)(a+b)$$

(9) Second symbol from the left hand is a

$$a(a+b)(a+b)^* (a+b) a(a+b)^*$$

(10) 10th symbol from right hand is a

$$(a+b)^* a(a+b)^9$$

(11) Third symbol from right end is a and fourth symbol is b

$$(a+b)^* b a(a+b)^*$$

note: when we need to remove complication from the end of the string add ϵ

- (12) To accept the words with two or more letters but beginning and ending with same letter when $\Sigma = \{a, b\}$

Case 1:-

$$a (a+b)^* a.$$

Case 2:-

$$b (a+b)^* b.$$

$$R.E \Rightarrow a (a+b)^* a + b (a+b)^* b$$

- (13) To accept the strings of 0's and 1's having no two consecutive zeros.

Case 1:- To accept the strings of the form 1^n or $0101 \dots$

$$R.E \quad 1^* + (01)^*$$

$$(1+01)^*$$

Case 2:- The above statement may end with zero or may not end with 0.

$$(1+01)^* (0+\epsilon)$$

- (14) To accept the strings of a's & b's whose length is either even or multiple of 3 or both

Case 1:- multiple of 2.

$$((a+b)(a+b))^*$$

Case 2:- multiple of 3

$$((a+b)(a+b)(a+b))^*$$

$$R.E \Rightarrow ((a+b)(a+b))^* + ((a+b)(a+b)(a+b))^*$$

- (15) Every block of 4 consecutive symbols contain at least two a's

$$(aa(a+b)(a+b) + a(a+b)a(a+b) + a(a+b)b(a+b)a + (a+b)aa(a+b) + (a+b)a(a+b)a + (a+b)(a+b)aa)^*$$

- (16) Obtain a R.E of the language $L = \{a^n b^m \mid m \text{ is even}\}$

Case 1:- If both n & m are even.

$$(aa)^*(bb)^*$$

Case 2:- If both n & m are odd.

$$a(aa)^* b(bb)^*$$

$$R.E \Rightarrow (aa)^*(bb)^* + a(aa)^* b(bb)^*$$

- (17) $L = \{a^n b^m \mid m \geq 1, n \geq 1, m \geq 3\}$

Case 1:- If $m=1, n \geq 3$.

$$(aaa)^* b$$

Case 2:- If $m \geq 1, m=3$

$$a(bbb)^* a b b b b^*$$

: If $n \geq 2, m \geq 2$.

$$aa a^* b b b b^*$$

$$R.E \quad aaaa^* b + a b b b b^* + a a a^* b b b b^*$$

- (18) $L = \{a^n b^m \mid n \geq 0, m \geq 0\}$

$$R.E$$

Note :- '0' symbol can be used in the place of '+'

even number of a's followed by even number of b's

$$(aa)^x(bb)^x$$

(19) strings of a's and b's containing not more than 3 a's on $\Sigma = \{a, b\}$

$$b^*(\epsilon+a)b^*(\epsilon+a)b^*(\epsilon+a)b^*$$

The string can have $0a, 1a, 2a, 3a$ i.e. $(\epsilon+a)(\epsilon+a)(\epsilon+a)$ we can insert any no. of b's anywhere in the string i.e. $b^*(\epsilon+a)b^*(\epsilon+a)b^*(\epsilon+a)b^*$

(20) $L = \{a^n b^m \mid n \geq 4, m \leq 3\}$

$$aaaaa^*(\epsilon+b)(\epsilon+b)(\epsilon+b)$$

$$(\epsilon+b)(\epsilon+b)(\epsilon+b)$$

(21) string length are multiples of 3 over $\Sigma = \{a, b\}$

$$(a^3b)^n \text{ mod } 3 = 0$$

$$((a+b)(a+b)(a+b))^*$$

(22) $L = \{w \mid n_0(w) \text{ mod } 3 = 0 \text{ where } \Sigma = \{a, b\}\}$

or

$$L = \{w \mid n_0(w) \text{ mod } 3 = 0\}$$

$$(b^*ab^*ab^*ab^*)^*$$

(23) set of strings do not end with 01 over $\Sigma = \{0, 1\}$

$$(0+1)^*(0+10+11)$$

(24) strings don't end with 110

$$(0+1)^*(000+010+100+111+001+011+101)$$

(25) no more than one b $\Sigma = \{a, b\}$

$$a^*(\epsilon+b)a^*$$

(26) no two consecutive letters are same. $\Sigma = \{a, b\}$

$$a^*ba^*$$

$$(\epsilon+b)(ab)^*(\epsilon+a)$$

$$(\epsilon+a)(\epsilon+b)^*a$$

$$a^*ba^*$$

$$(\epsilon+a)(ba)^*(\epsilon+b)$$

$$a^* \rightarrow (\epsilon+b)(ab)^*(\epsilon+a) + (\epsilon+a)(ba)^*(\epsilon+b)$$

(27) write RE for the following $\Sigma = \{a, b, c\}$

a) containing exactly one a.

b) no more than 30's

c) at least one occurrence of each symbol in Σ .

(28) every 'a' in w is immediately preceded and followed by b $\Sigma = \{a, b\}$

(29) w does not end in 'ba' on $\Sigma = \{a, b\}$

(30) w is a odd number without trailing b's

31) w has both aa and bb as substring
 32) w contains at least 2 b's that are not followed by an a .

33) $L = \{ a^n b^m c^p \mid n \leq 4, m \geq 2, p \leq 23 \}$
 34)

37) a) $(b+c)^* a (b+c)^*$
 b) $(b+c)^* (c+a) (b+c)^* (c+c) (b+c)^* c$

38) $bab (bab)^*$

39) $(a+b)^* (aa+ab+bb)$

30)

31) $(a+b)^* (aa) (a+b)^* (bb) (a+b)^* + (a+b)^* bb (a+b)^* aa (a+b)^*$

32) $a^* b b b a^*$

33) $b^* (c+a) b^* (c+a) b^* (c+a) b^*$

34) $(c+a) (c+a) (c+a) (c+a) b b b^* (c+c) (c+c)$

* Relation between finite automata and regular expression.

Every language defined by a regular expression is defined by one of these automata.

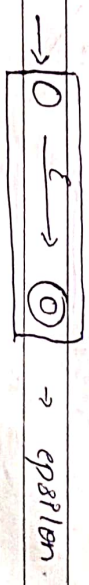


Proof:-

$L = L(\emptyset)$ for regular expression $\emptyset$. We show that $L = L(\emptyset)$ for some ϵ -NFA with:

- 1) Exactly one accepting state.
- 2) No more than the initial state.
- 3) No more out of the accepting state.

Basis: $\emptyset, \epsilon, a$ are all regular expression of the language $\emptyset, \epsilon, a$ which are ϵ respectively. The automaton corresponding to these expression are.



→ problem.



→ empty language.



Induction: Now we work with the automata for regular expression $R+S$, $R.S$, R^* . when R & S are regular expressions constructed from primitive regular expressions.

$R+S$: Union.

$R.S$: Concatenation.

R^* : Kleene star.

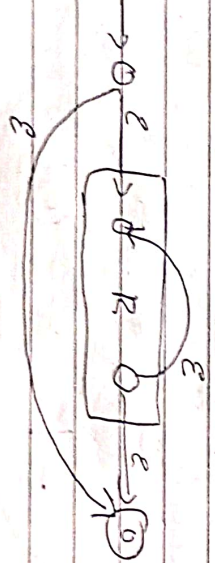
Case 1: Automata for $R+S$



Case 2: Automata for $R.S$



Case 3: Automata for R^* .



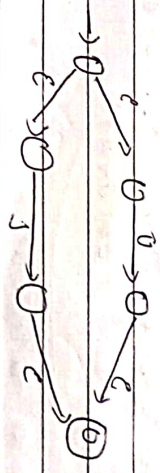
① Convert the regular expression $(0+1)^*1(0+1)$ to an ϵ -NFA.

⇒ Subexpression are: 0, 1, $(0+1)$, $(0+1)^*$, $(0+1)^*1$, $(0+1)^*1(0+1)$

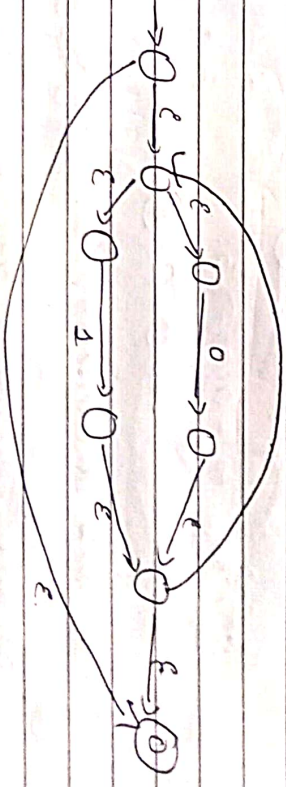
i] NFA to accept 0: → 0 → 1

ii] NFA to accept 1: → 0 → 1 → 1

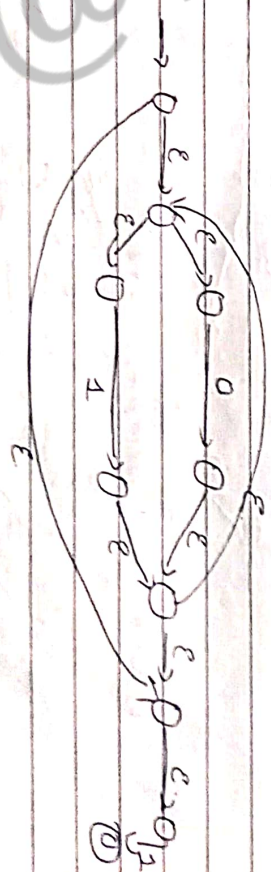
iii] NFA to accept 0+1:-



iv] NFA to accept $(0+1)^*$.



v] NFA to accept $(0+1)^*1$



vi] NFA to accept $(0+1)^*1(0+1)$



③ Convert the regular expression $ab(a+b)^*$ to an NFA

→ Subexpressions are: $a, b, (a+b), (a+b)^*, ab(a+b)^*$

i] NFA to accept a :

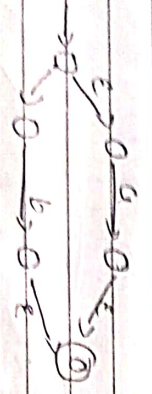


ii] NFA to accept b :

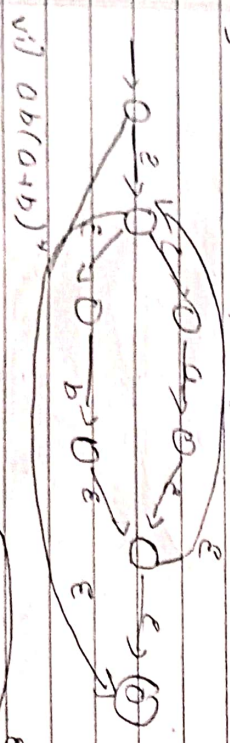


iii] NFA to accept $a+b$:

iv] NFA to accept $a+b$



v] NFA to accept $(a+b)^*$



④ $a^* + b^* + c^*$

→ Subexpression are: $a, b, c, a^*, b^*, c^*, a^* + b^* + c^*$

i] NFA to accept a :

ii] NFA to accept b :

iii] NFA to accept c :

iv] NFA to accept a^* :



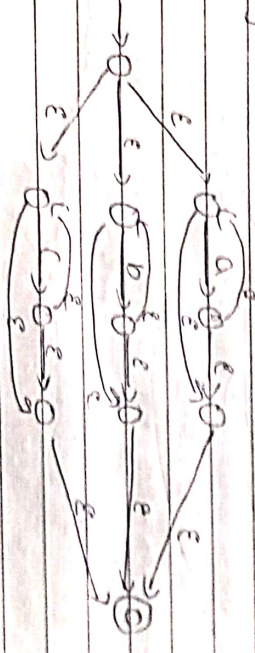
v] NFA to accept b^* :



vi] NFA to accept c^* :



vii] $a^* + b^* + c^*$



⑤ $(a+b)^* a (a+b)^*$

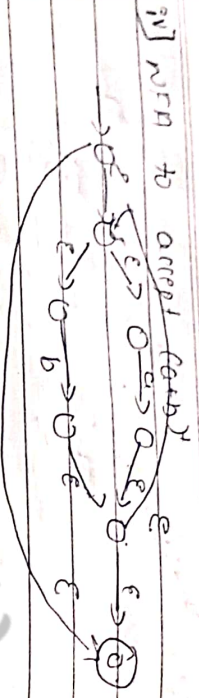
→ Subexpressions are: $a, b, (a+b), (a+b)^*, a(a+b)^*, (a+b)^* a$

i] NFA to accept a :

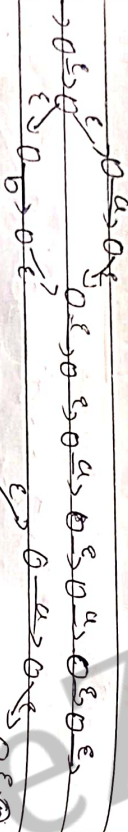
ii] NFA to accept b :

iii] NFA to accept $a+b$:



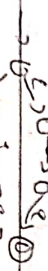


iv] NFA to accept $(a+b)^*$
 i] NFA to accept a^* : $\rightarrow q_0 \xrightarrow{a} q_1 \xrightarrow{\epsilon} q_0$
 ii] NFA to accept $(a+b)^* a a$:
 iii] NFA to accept $(a+b)^* a a (a+b)^*$

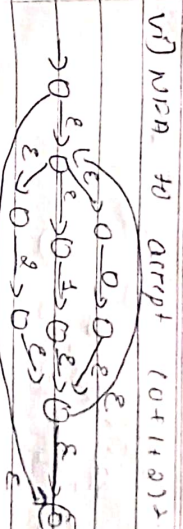
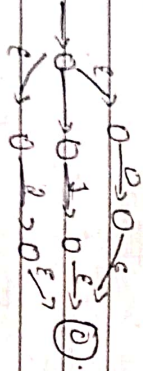


⑤ $(0+1+2)^* 0 (0+1)$

Subexpression are: $0, 1, 2, 0+1, 0+1+2, (0+1+2)^*, 0(0+1)$
 i] NFA to accept 0 : $\rightarrow q_0 \xrightarrow{0} q_1$
 ii] NFA to accept 1 : $\rightarrow q_0 \xrightarrow{1} q_1$
 iii] NFA to accept 2 : $\rightarrow q_0 \xrightarrow{2} q_1$
 iv] NFA to accept $0+1$:
 v] NFA to accept $0+1+2$:
 vi] NFA to accept $0(0+1)$:



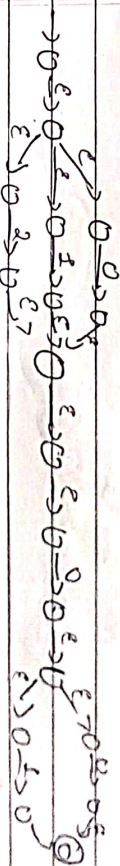
v] NFA to accept $0(0+1)$:



vi] $0(0+1)$



vii] NFA to accept $(0+1+2)^* 0(0+1)$



* Converting DFA's to Regular expression by eliminating states

1 Remove unreachable states from N.

* If N has no-accepting states then return ϕ

* If the start state of N is part of a loop, create a new start states and connect to the start state via an ϵ -transition.

* If there is more than one accepting state of N or there is any transition out of any of them create a new accepting state & create each of the states to it via an ϵ -transition. The old accepting states are no longer accepting.

* Until only the start state and accepting state remain do:

i] select np state.

ii] Remove np state. for N.

iii] modify the transitions among the remaining states so as to accept the same strings

* Return the regular expression that labels the one remaining transition from the start state to the accepting state

iv] Return the regular expression that labels the one remaining transition from the start state to the accepting state

v] Return the regular expression that labels the one remaining transition from the start state to the accepting state

vi] Return the regular expression that labels the one remaining transition from the start state to the accepting state

1 Autom D.F for the following automata

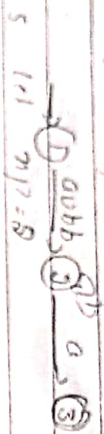


2: No unreachable state.

3: No new start state is required.

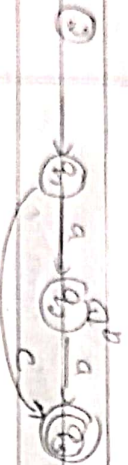
4: No new final state is required.

5: Let $np = q_1$



6: Let $np = q_1$

7: Let $np = q_1$



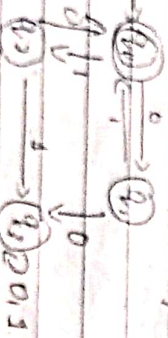
8: No new & final state is required.

9: Let $np = q_1$



10: Let $np = q_1$

11: Convert the following finite automata to R.C.



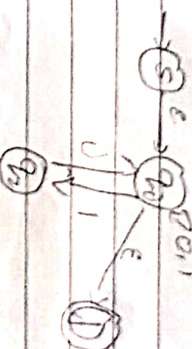
12: Remove dead state q_2 .



2.



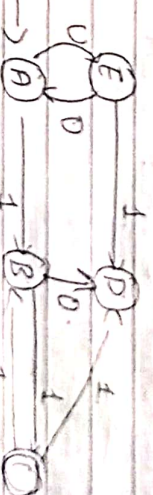
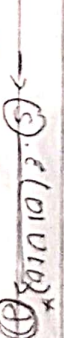
3: Let $np = q_1$



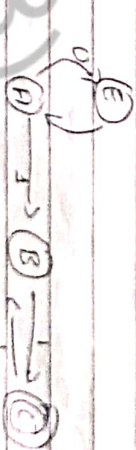
4: Let $np = q_1$



5: Let $np = q_1$



6: Remove dead state D



7: new start & final state is required



8: Let $np = q_1$



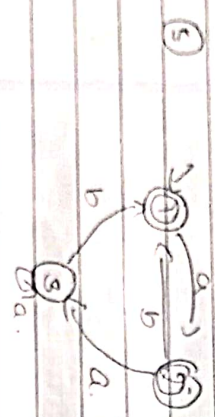
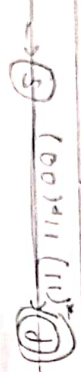
1: 1st NP = A



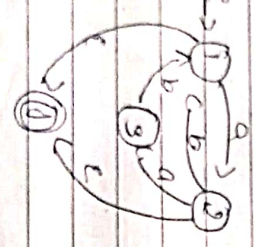
2: 1st NP = B



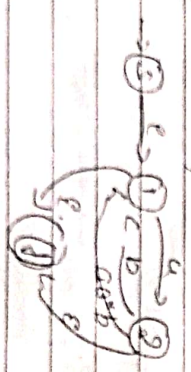
3: 1st NP = C



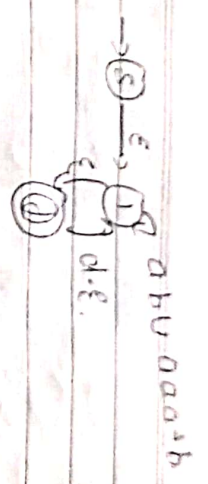
1: 1st NP = A and 2nd NP = B



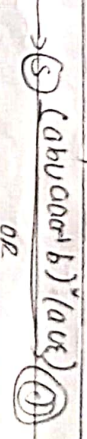
2: 1st NP = A



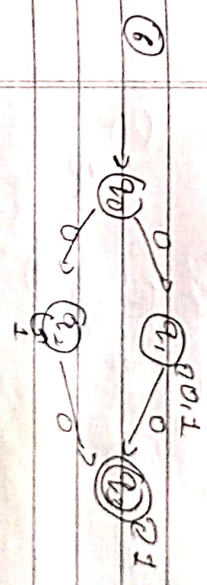
3: 1st NP = A



1: 1st NP = A



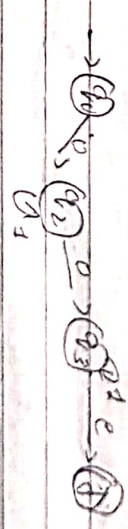
2: 1st NP = A and 2nd NP = B



1: 1st NP = A and 2nd NP = B



2: 1st NP = A and 2nd NP = B



3: 1st NP = A



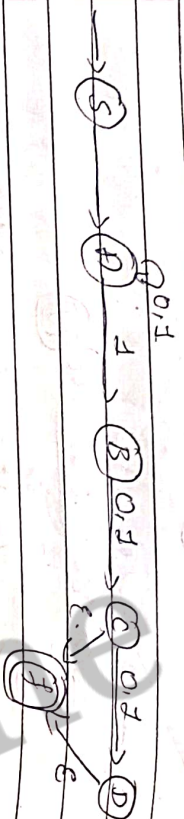
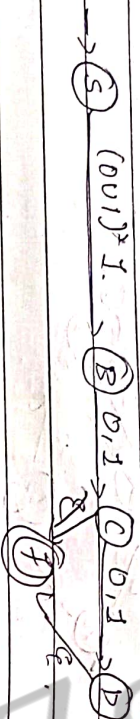
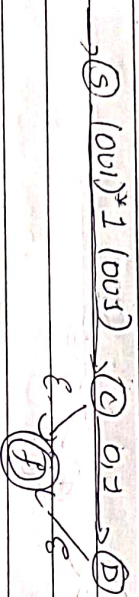
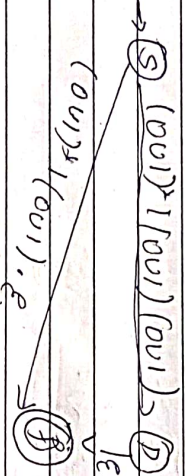
4: 1st NP = A



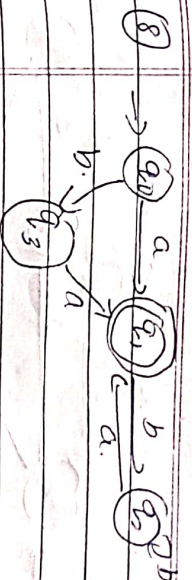
5: 1st NP = A



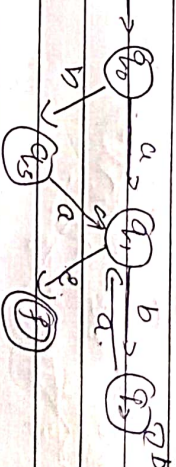
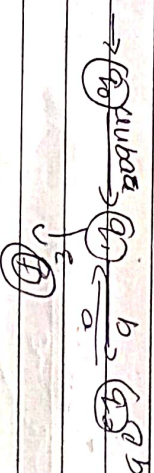
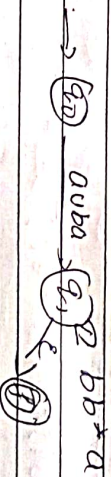
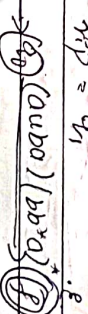
1: NO unreachable state.
2: new state and final state is required.


$$\underline{3. \text{rip} = A.}$$

$$A \quad r^2 p = B$$

$$5 \pi p = c$$

$$6. \quad \text{HP} = D \quad (OU1)^x, (OU1)(OU1)^2$$

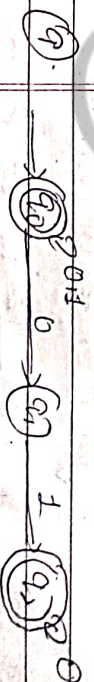
$$\begin{aligned} R \cdot f &:= (Dv)^x \mid (v \mid) \cup (v \mid)^x \mid (v \mid)(v \mid) \\ R \cdot f &\rightarrow (Dv)^x \mid (v \mid) (f \cdot (v \mid)) \end{aligned}$$



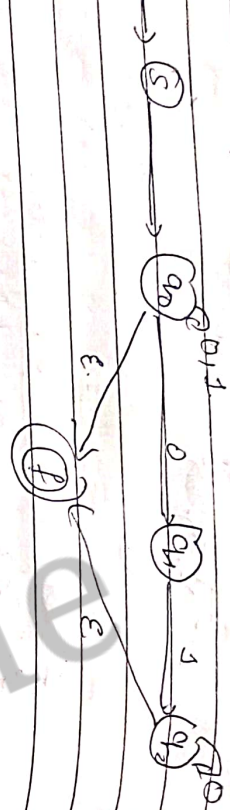
1: no unreachable state.
2: new final state.


$$3: \text{app} = a_3.$$
4. $r'p = qz.$ 
$$S: \mathcal{M}_P = \mathcal{M}_1$$


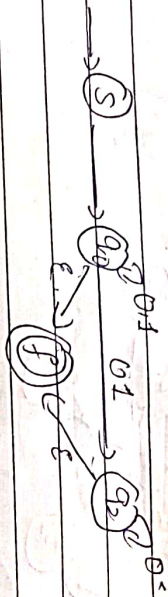
$$S_{\mathbb{R}G} = (auba)(bb^*a)^*$$



1: new final state i.e start state.



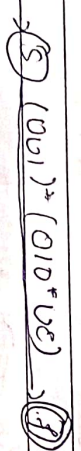
2: $r.p = 01$



3: $r.p = q_2$



4: $r.p = q_0$



D.C = $(001)^* (010^* 01)^*$



- 1) no reachable state.
- 2) new start & final state is required.



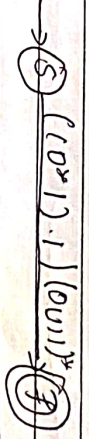
$r.p = q_0$



$r.p = q_1$



$r.p = q_2$

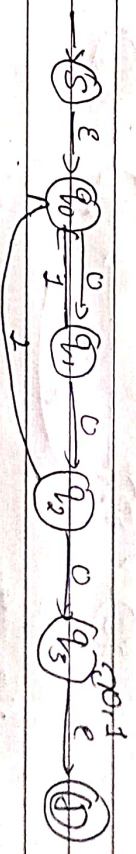


D.C = $(10^* 1)^* (10011)^*$

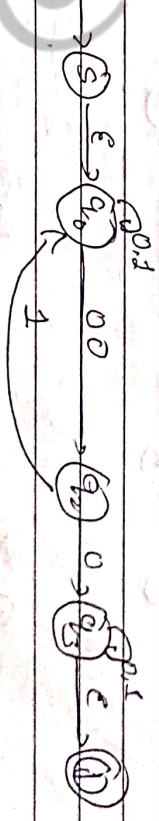


- 1) No unreachable states.
- 2) New start & final state is required.

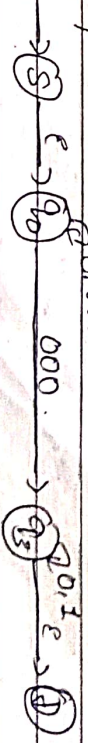
3) New start & final state is required.



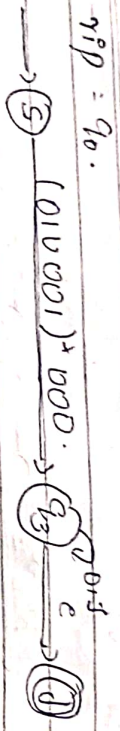
$r.p = q_1$



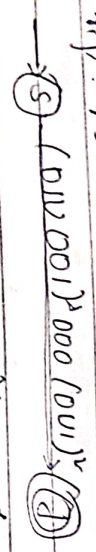
$r.p = q_2$



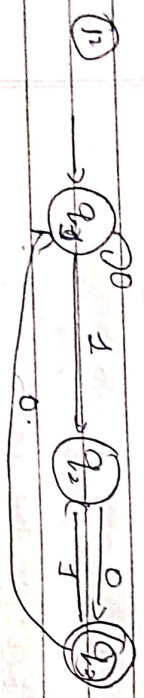
$r.p. = q_0.$



$r.p. = q_3.$

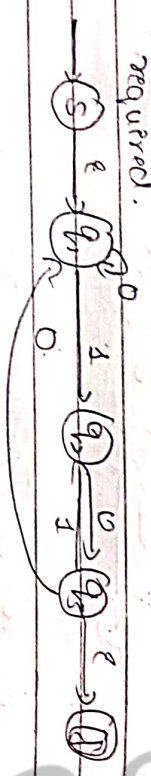


$R.E = (010001)^* 0001001)^*$

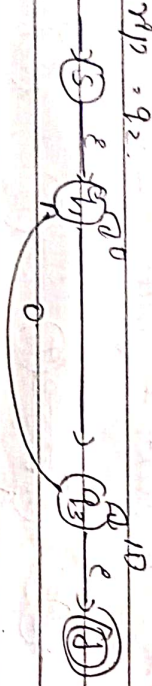


① No unreachable state.

② new start state & final state are required.



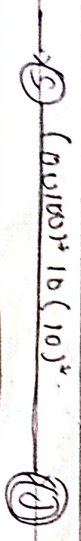
$r.p. = q_2.$



$r.p. = q_3.$



$r.p. = q_1.$



$R.E = (00100)^* 10(10)^*$

⇒ Pumping lemma for regular languages

- ① $|xy| \leq k$
- ② $y \neq \epsilon$ i.e. $|y| > 0$ (i.e. y is non empty)
- ③ for all $q \geq 0, xy^qz \in L$.

① Show that 10^m0^n is not regular.

proof:-

Assume L is regular

let $w = 0^k10^k$, a long string

∴ length of $w = 2k$.

$w = xy^2z$. should satisfy

pumping lemma condition.

① $|xy| \leq k$, q_1 point $xy = a^p$

② $y \neq \epsilon$, $p > 0$

③ $q \geq 0, xy^qz \in L$.

let $k=3, w = 0001000 \in L$.

$x = a$

$y = 0a$

then according to lemma's rule.

$xy^2z \in L$ for $q=2$.

$xy^2z = xy^2z \cdot a(00)^2 bbb$

$000000bbb$
 $05b^3 \notin L$
 $0^k b^k \notin L$

which is a contradiction for our assumption $\therefore L$ is not a regular language.

② $L = \{w = f^k, (g^* : \text{this part has even no. balanced})\}$

Assume L is regular.

Let $w = f^k$ be a long string

$\therefore w = g^k$ should satisfy pumping lemma conditions.

① $|xy| \leq k \Rightarrow \therefore xy$ is in first a^k part

② $y \neq \emptyset$ as $|y| > 0$ (non-empty)

③ for each $q \geq 0, xy^qz \in L$.

Let $f = y$ $w = f^k = (f^q)^q \in L$ where $x = a^k$ $z = f^k$

According to the rule

$$xy^qz \in L \text{ for } q = 2$$

$$xy^2z = f^2 f^k f^k = f^{2k+2} \notin L$$

③ Show that even palindromic language is not regular $L = \{w = a^k b^k\}$

Let L be the regular language and $w = a^k b^k$ are long string

$\therefore$ length of $w = 4k \leq w = xy^2z$ should satisfy pumping lemma conditions.

i) $|xy| \leq k$

$\therefore xy$ is in first a^k part & $y = a^q$ for some

ii) $y \neq \emptyset$ $y \neq \emptyset$

$\therefore p > 0$.
iii) $Aq \geq 0$ $xy^qz \in L$

Let $k = 3$.

$$w = a^3 b^3 a^3$$

$$aaq bbb bbaa$$

$$xy = z$$

$$x = a$$

$$y = aa$$

$$z = bbb bbaa$$

The according to pumping lemma's third rule $xy^2z \in L$ for $q = 2$. i.e.

$$xy^2z = a(aa)^2 bbb bbaa$$

$$= a a a a a b b b b b a a a$$

$$= a^5 b^3 a^3$$

$$= a^5 b^3 b^3 a^3 \notin L$$

$$= a^5 b^3 b^3 a^3 \notin L$$

$\therefore$ We got a contradiction to our assumption hence it is not regular.

④ Show that $L = \{a^n b^n : n \geq m\}$ is not regular

Assume L is a regular language.

$w = a^k b^k$ are long string

$\therefore$ length of $w = 2k+1$ & $w = xy^2z$ should satisfy pumping lemma condition.

i) $|xy| \leq k$

$\therefore xy$ is in first a^k part & $y = a^q$ for some p

if $y \neq \emptyset \therefore p > 0$
 iii) $xqz \in L$

$\therefore$ Those are already more no. of a 's than b 's so if we pump in the no. of a 's the resulting language will be regular

$\therefore$ we should pump out a 's. i.e. let $q = 0$.

let $k = 3, w = a^{3+1} b^3$
 $\therefore a^4 b^3$

$\therefore a^4 a^0 b^3 b^3$

xqz

$x = aa$

$y = a$

$z = abbb$

The according to third rule $xy^kz =$

$xy^0z = aa(a)^0 abbb$

$\therefore aabbbb$

$\therefore a^0 b^4 \notin L$

$\therefore$ We got a contradiction for language

$\therefore$ it is not a regular language.

⑤ $\{w \in \{a,b\}^* : w \text{ contains exactly two more } b\text{'s than } a\text{'s}\}$

$w = a^k b^{k+2}$ a long string

$\therefore |w| = 2k + 2, w = xy^2z$

① $|xy| \leq k \therefore xy$ is in first a^k

⑥ $y \in \emptyset$ & $|y| > 0 \& p > 0$

⑦ for all $q \geq 0, xy^qz \in L$
 $x = a, y = aa, z = bbbbbb$

$xy^0z \in L$ for $q = 2$

$xy^2z = a(aaa)^2 bbbbbb$

$\therefore a^5 b^7$

$\therefore a^n b^{n+2} \notin L$

eg.

we got contradiction for our assumption

Therefore it is not regular expression.

⑧ $pw \in \{0,1\}^* : \exists x \in \{0,1\}^+ (w = xxx^R x)$

$w = a^k a^l a^k b$ is a long string

$w = 3k + 3, w = xy^2z$ should satisfy.

① $|xy| \leq k \therefore xy$ is in 1^{st} a^k part

$y = a^2$

② $xy \in \emptyset$ or $|y| > 0 \& p > 0$.

③ for all $q \geq 0, xy^qz \in L$

let $k = 5$

$w = a^5 b b a^5 a^5 b$

$x = a, y = aa, z = b b a a a a a b$

according to the pumping lemma 3rd rule

$xy^qz =$ for $q = 2$

$xy^2z = a(aaa)^2 (bbaaaaaab)$

$\therefore a^5 b b a^5 a^5 b \notin L$

7) Show that $L = \{a^n : n \text{ is prime number}\}$ is not regular.

let $w = a^p$ is a lang string
 $\therefore$ length $w = p$ $w = xyz$ should satisfy

① $|xy| \leq p$ $\therefore xy$ is least in p part

② $y \neq \emptyset$ $\therefore p > 0$

③ for $q > 0$ $xq^2z \in L$

now let $k = 5$

$w = a^5 = aaaaa$ but $x = aaaa$ $y = a$ $z = \epsilon$

$q = 2$

$xy^2z = aaaa(a)^2a$

$= aaaaaaa \notin L$

$\therefore$ it is not a regular language

2) Properties of regular languages.

① Closure properties.

$\rightarrow$ Union

$\rightarrow$ Intersection

$\rightarrow$ Complement

$\rightarrow$ Difference

$\rightarrow$ reversal

$\rightarrow$ closure.

$\rightarrow$ concatenation.

$\rightarrow$ homomorphism

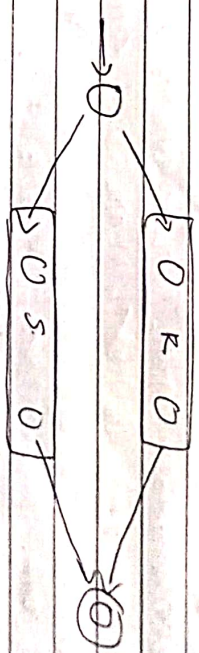
$\rightarrow$ inverse homomorphism

3) Union, concatenation & Kleene star (*)

Theorem:- The regular languages are closed under union and Kleene star.

Proof:- Union.

Since L and M are regular, they have regular expressions say $L = L(R)$ and $M = L(S)$. Then $L \cup M = L(R \cup S)$ by the definition of the $\cup$ operator for regular expression.



concatenation.

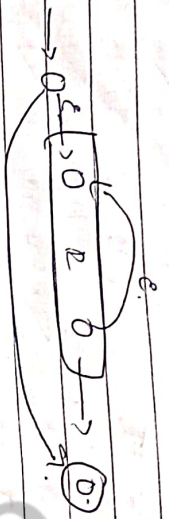
Since L & M are regular they have regular expressions say $L = L(R)$ & $M = L(S)$. Then $LM = L(RS)$ by the defn of the $\cdot$ operator for regular expression.



closure (Kleene star)

Since L are regular they have regular expressions say $L = L(R)$ by the defn of the $*$ operator for regular

expression.



→ Equivalence and minimization of automata.

→ Equivalence class.

→ two states are equivalent / indistinguishable if and only if both p & q are final or both of them are not final.

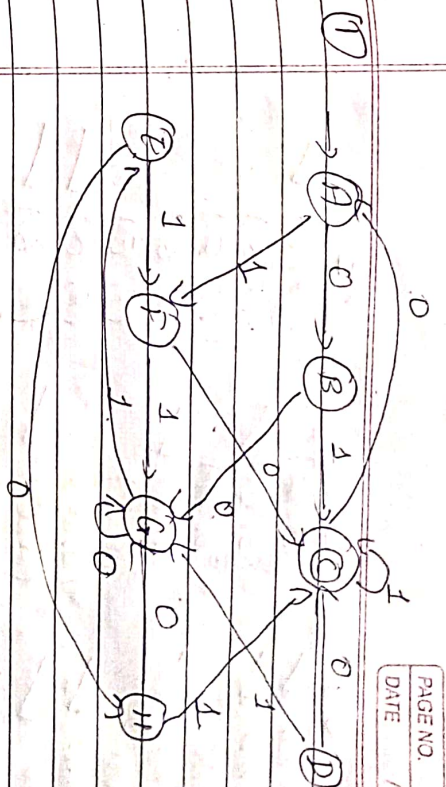
→ Distinguishable states / indistinguishable states:-

→ If two states are not equivalent then we say they are distinguishable. Test on state p in distinguishable from state q if there is atleast one string to such that one of the $\delta(p, w)$ & $\delta(q, w)$ is accepting and other is not accepting.

* Table-filling algorithm / minize the DFA

Steps:-

- ① Finding distinguishable and indistinguishable states using table filling algorithm.
- ② Obtain the states of minimized DFA using groups
- ③ Compute the transition table.
- ④ Identify start & final states
- ⑤ Draw minimized DFA.



② a) obtain the transition table for DFA.

	0	1
0	0	1
1	3	4
2	5	6
3	6	4
4	6	6
5	6	6
6	6	6

③ Construct table for markings.

	0	1
0	*	*
1	*	*
2	*	*
3	*	*
4	*	*
5	*	*
6	*	*

δ	0	1
AF $\checkmark$	(B, G)	(F, C) $\checkmark$
AD $\checkmark$	(B, C) $\checkmark$	(F, G)
AF $\checkmark$	(B, H)	(F, F)
AF $\checkmark$	(B, C) $\checkmark$	(F, G)
AG	(B, G)	(F, F)
AF $\checkmark$	(B, G)	(F, C) $\checkmark$
BD $\checkmark$	(G, C) $\checkmark$	(G, G) $\checkmark$
BE $\checkmark$	(G, H)	(G, F) $\checkmark$
BF $\checkmark$	(G, C) $\checkmark$	(G, G) $\checkmark$
BG $\checkmark$	(G, G)	(G, F) $\checkmark$
BH	(G, G)	(G, C) $\checkmark$
DE $\checkmark$	(G, H) $\checkmark$	(G, F)
DF	(G, C)	(G, G)
DG $\checkmark$	(G, G) $\checkmark$	(G, F)
DH $\checkmark$	(G, G) $\checkmark$	(G, C)
EF $\checkmark$	(H, C) $\checkmark$	(F, G)
EG	(H, G)	(F, F)
EH $\checkmark$	(H, G)	(F, C) $\checkmark$
FG $\checkmark$	(C, G) $\checkmark$	(G, F)
FH $\checkmark$	(C, G) $\checkmark$	(G, C) $\checkmark$
GH $\checkmark$	(G, G)	(G, C) $\checkmark$

rewrite the above table for finding

distinguishable & undistinguishable states.

HE	(B, H)	(F, F)
AG $\checkmark$	(B, G) $\checkmark$	(F, G)
BH	(G, G)	(C, C)
DE	(C, C)	(G, G)
EG $\checkmark$	(H, G) $\checkmark$	(F, F)

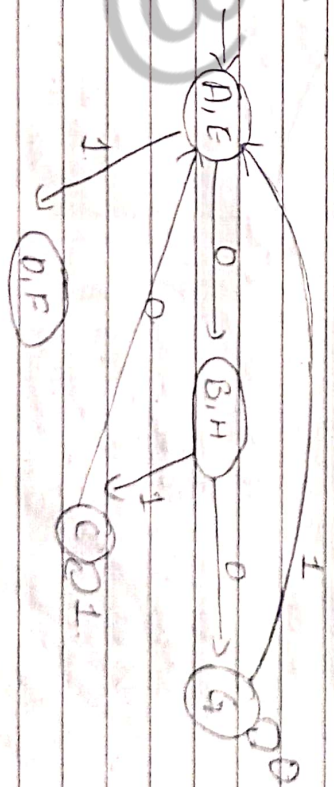
Identically indistinguishable pairs:-

(D, E) (B, H) (D, F)

Distinguishable pairs:-
C and G.

(3) Transition table.

δ	0	1
$\rightarrow$ (A, E)	(B, H)	(F, F) replace (D, F)
(B, H)	G	C
(D, F)	C	G
$\times$ C	A (D, E)	C
G	G	E replace (D, E)



15

	0	1
→ A	B	A
B	A	C
C	D	B
*D	D	A
E	D	E
F	G	E
G	F	G
H	G	D

B								
C								
D	*		*	*				
E					*			
F					*			
G					*			
H					*			

501.

(A, B)	B	A
(A, C)	B	B
(A, E)	B	F
(B, F)	B	E
(B, G)	B	G
(B, H)	A	G

$(1,1)$	D	F
$(1,2)$	D	E
$(1,3)$	D	G
$(1,4)$	D	G
$(2,1)$	D	E
$(2,2)$	D	G
$(2,3)$	D	G
$(2,4)$	F	G

D. J.

(A, B)	BA	AC
(A, C)	BD	AB
(A, E)	BD	AF
(A, F)	BG	AE
(A, G)	BC	AG
(A, H)	BG	AD
(B, C)	AD	CB
(B, E)	AD	CF
(B, F)	AG	CE
(B, G)	AF	CG
(B, H)	AG	CD
(C, E)	DD	BC
(C, F)	DB	BC
(C, G)	DF	BG
(C, H)	DB	BD
(E, F)	DG	FE
(E, G)	DF	FG
(E, H)	DB	FD
(F, G)	GF	EG
(F, H)	GG	ED
(G, H)	GG	GD

[illegible]

δ	θ	γ
$(A, B) \checkmark$	(B, A)	$(A, C) \checkmark$
$(A, E) \checkmark$	(B, G)	$(A, E) \checkmark$
(A, G)	(B, F)	(A, G)
(B, F)	(A, G)	(C, E)
(B, G) $\checkmark$	(A, E)	(C, G)
(C, E)	(D, D)	(B, F)
$(F, G) \checkmark$	(G, F)	(E, G)

⇒ Indisfingierbar :-

 $(A, h) \quad (B, F), \quad (C, E)$

deftinguishable.

D
A
H

⇒ Transmission tab 12.

δ	0	1
$\rightarrow (A, b)$	(B, F)	(A, b)
(B, F)	(A, b)	(C, E)
(C, E)	D	(B, F)
X D	D	(A, b)
H.	(A, b)	D.

	D	B	C	F	H
$\rightarrow D$	B	C	F	H	I
B	C	F	H	I	J
C	F	H	I	J	K
F	H	I	J	K	L
H	I	J	K	L	M
I	J	K	L	M	N
J	K	L	M	N	O
K	L	M	N	O	P
L	M	N	O	P	Q
M	N	O	P	Q	R
N	O	P	Q	R	S
O	P	Q	R	S	T
P	Q	R	S	T	U
Q	R	S	T	U	V
R	S	T	U	V	W
S	T	U	V	W	X
T	U	V	W	X	Y
U	V	W	X	Y	Z
V	W	X	Y	Z	A
W	X	Y	Z	A	B
X	Y	Z	A	B	C
Y	Z	A	B	C	D
Z	A	B	C	D	E
A	B	C	D	E	F
B	C	D	E	F	G
C	D	E	F	G	H
D	E	F	G	H	I
E	F	G	H	I	J
F	G	H	I	J	K
G	H	I	J	K	L
H	I	J	K	L	M
I	J	K	L	M	N
J	K	L	M	N	O
K	L	M	N	O	P
L	M	N	O	P	Q
M	N	O	P	Q	R
N	O	P	Q	R	S
O	P	Q	R	S	T
P	Q	R	S	T	U
Q	R	S	T	U	V
R	S	T	U	V	W
S	T	U	V	W	X
T	U	V	W	X	Y
U	V	W	X	Y	Z
V	W	X	Y	Z	A
W	X	Y	Z	A	B
X	Y	Z	A	B	C
Y	Z	A	B	C	D
Z	A	B	C	D	E
A	B	C	D	E	F
B	C	D	E	F	G
C	D	E	F	G	H
D	E	F	G	H	I
E	F	G	H	I	J
F	G	H	I	J	K
G	H	I	J	K	L
H	I	J	K	L	M
I	J	K	L	M	N
J	K	L	M	N	O
K	L	M	N	O	P
L	M	N	O	P	Q
M	N	O	P	Q	R
N	O	P	Q	R	S
O	P	Q	R	S	T
P	Q	R	S	T	U
Q	R	S	T	U	V
R	S	T	U	V	W
S	T	U	V	W	X
T	U	V	W	X	Y
U	V	W	X	Y	Z
V	W	X	Y	Z	A
W	X	Y	Z	A	B
X	Y	Z	A	B	C
Y	Z	A	B	C	D
Z	A	B	C	D	E
A	B	C	D	E	F
B	C	D	E	F	G
C	D	E	F	G	H
D	E	F	G	H	I
E	F	G	H	I	J
F	G	H	I	J	K
G	H	I	J	K	L
H	I	J	K	L	M
I	J	K	L	M	N
J	K	L	M	N	O
K	L	M	N	O	P
L	M	N	O	P	Q
M	N	O	P	Q	R
N	O	P	Q	R	S
O	P	Q	R	S	T
P	Q	R	S	T	U
Q	R	S	T	U	V

B	*		*								
C	*		*								
D	.			*							
E	*		*		* ✱						
F	*	*		.	*						
G				*		*					
H	.		*				*				
J	*	*		*	*	*	*	*	*	*	*

J	②	0	I.
AB ✓	BF	BC ✓	EH
AD	BH	BE	FI
AE ✓	BI.	BF ✓	EJ
AG	B J 3	BH.	EB
AH ✓	BI	BI ✓	EI
BD ✓		CF ✓	FH
BE		CF	FI
BG ✓		CH ✓	FB
BH.		CL	FI
CF		DG	HB

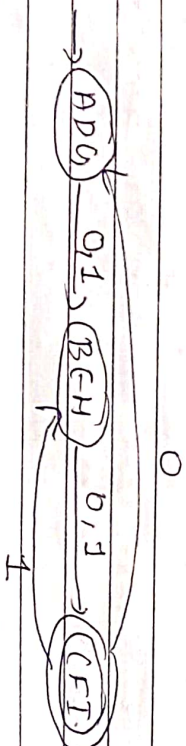
hamilton tabel

907

$$\rightarrow (ADG) \quad (BEH) \quad (BFH)$$

$(B \in H) \quad (I \neq J) \quad (I \neq J)$

8. $LCPI$ (ADG) $\cdot$ (BEH)



① obtain the R.F. for the following csm.



$$1^* 0 \cup 1^* (01)^+ j^* j \cup 1^* (01)^* 1^* (b+1)^+$$

Indistinguishable states

(A, D)	(A, G)	(B, E)	(B, H)	(C, F)	(C, I)
----------	----------	----------	----------	----------	----------

$$(0,6) \quad (F, H) \quad (F, HI)$$
 $(A, D) (A, b) (D, b) \rightarrow 1 \text{ group}$
$$(B, E) \quad (B, H) \quad (E, H) \rightarrow 2 \text{ groups}$$

$(EF) \quad (IT) \quad (FJ) \rightarrow S$ group

clinfringibile.

no distinguishing state.